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JEE MAIN PHYSICS DPP: LEAST COUNT & SIGNIFICANT FIGURES

Topic: Vernier Calipers, Screw Gauge, Spherometer & Significant Figures | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A vernier caliper has 1 mm marks on the main scale. It has 20 equal divisions on the vernier scale which match with 16 main scale divisions. The least count of this caliper is:
- 0.02 mm
 - 0.05 mm
 - 0.2 mm
 - 0.1 mm
- Q2.** A screw gauge has a pitch of 0.5 mm and a circular scale divided into 50 divisions. When the two studs are in contact without any object, the 45th division of the circular scale coincides with the reference line and the zero of the main scale is barely visible. The zero error of the instrument is:
- +0.05 mm
 - 0.05 mm
 - 0.45 mm
 - +0.45 mm
- Q3.** The number of significant figures in the numbers 0.003020, 4.500×10^3 , and 200.0 are respectively:
- 4, 4, 4
 - 3, 2, 1
 - 4, 2, 4
 - 6, 4, 1
- Q4.** In a vernier caliper, N divisions of the vernier scale coincide with $(N - 1)$ divisions of the main scale. If 1 MSD = a units, the least count of the vernier caliper is:
- $\frac{a}{N}$
 - $\frac{a}{N - 1}$
 - $\frac{a}{N + 1}$
 - $\frac{(N - 1)a}{N}$
- Q5.** The pitch of a screw gauge is 1 mm and there are 100 divisions on its circular scale. When measured, the main scale reads 2 mm and the 52nd division of the circular scale coincides with the reference line. If the screw gauge has a negative zero error of 0.03 mm, the corrected diameter of the wire is:
- 2.49 mm
 - 2.52 mm
 - 2.55 mm
 - 2.58 mm
- Q6.** Solve the following expression to appropriate significant figures:
- $$4.28 - 0.125 + 1.2$$
- 5.355
 - 5.36
 - 5.4
 - 5.35
- Q7.** The length and breadth of a thin rectangular metal plate are measured using a vernier caliper to be $l = 15.42$ cm and $b = 4.2$ cm. The area of the plate reported up to correct significant figures is:
- 64.764 cm²
 - 64.76 cm²
 - 64.8 cm²
 - 65 cm²
- Q8.** A spherometer has a pitch of 0.5 mm and a circular disc divided into 100 divisions. The least count of the spherometer is:
- 0.005 mm
 - 0.001 mm
 - 0.05 mm
 - 0.0005 mm
- Q9.** In a travelling microscope, the main scale has 0.5 mm divisions. The vernier scale has 50 divisions which coincide with 49 main scale divisions. The least count of the travelling microscope is:
- 0.01 mm
 - 0.001 mm
 - 0.02 mm
 - 0.005 mm
- Q10.** A screw gauge gives the following reading when used to measure the thickness of a glass sheet: Main scale reading = 3 mm, circular scale reading = 68 divisions. Given that 1 mm on the main scale corresponds to 100 divisions of the circular scale and the instrument has a positive zero error of +0.04 mm, the actual thickness of the sheet is:
- 3.64 mm
 - 3.68 mm
 - 3.72 mm
 - 3.60 mm
- Q11.** The side of a solid cube is measured as $x = (2.50 \pm 0.01)$ cm. The percentage error in the measurement of its volume is:

- (a) 0.4%
(b) 0.8%
(c) 1.2%
(d) 1.6%
- Q12.** In a specially designed vernier caliper, each 1 cm on the main scale is divided into 10 equal parts. 10 divisions of the vernier scale coincide with 11 divisions of the main scale. The least count of this instrument is:
(a) 0.1 mm
(b) 0.01 mm
(c) 0.11 mm
(d) 0.09 mm
- Q13.** Round off the number 3.745 and 3.735 to three significant figures according to standard scientific rounding rules:
(a) 3.75 and 3.74
(b) 3.74 and 3.74
(c) 3.75 and 3.73
(d) 3.74 and 3.73
- Q14.** The mass and volume of a body are measured as $m = 24.2 \text{ g}$ and $V = 4.7 \text{ cm}^3$. The density of the body reported to appropriate significant figures is:
(a) $5.1489 \text{ g} \cdot \text{cm}^{-3}$
(b) $5.15 \text{ g} \cdot \text{cm}^{-3}$
(c) $5.1 \text{ g} \cdot \text{cm}^{-3}$
(d) $5.2 \text{ g} \cdot \text{cm}^{-3}$
- Q15.** When measuring the internal diameter of a cylinder using a vernier caliper with least count 0.01 cm, the zero of the vernier scale lies between 3.2 cm and 3.3 cm of the main scale, and the 4th vernier division coincides with a main scale division. If the caliper has a negative zero error of -0.02 cm , the correct diameter is:
(a) 3.22 cm
(b) 3.24 cm
(c) 3.26 cm
(d) 3.28 cm
- Q16.** If 1 MSD = 0.5 mm in a vernier caliper and 50 vernier scale divisions are equal to 49 main scale divisions, how many vernier divisions correspond to a measurement difference of 0.14 mm?
(a) 7
(b) 14
(c) 28
(d) 35
- Q17.** The diameter of a cylinder is measured with a screw gauge having 50 divisions on circular scale and pitch 0.5 mm. If the student takes 5 complete rotations of the screw to move through the thickness of 2.5 mm, the least count and the pitch respectively are:
(a) 0.01 mm, 0.5 mm
(b) 0.005 mm, 0.5 mm
(c) 0.01 mm, 1.0 mm
(d) 0.02 mm, 0.5 mm
- Q18.** What is the sum of $3.0 \times 10^{-3} \text{ kg}$ and $4.50 \times 10^{-2} \text{ kg}$ expressed in proper significant figures?
(a) $4.80 \times 10^{-2} \text{ kg}$
(b) $4.8 \times 10^{-2} \text{ kg}$
(c) $48.0 \times 10^{-3} \text{ kg}$
(d) 0.048 kg
- Q19.** The main scale of a vernier caliper reads in millimetres. 10 divisions of the vernier scale coincide with 9 divisions of the main scale. While measuring the thickness of a coin, the main scale reads 2 mm and the 7th vernier division coincides with a main scale mark. The thickness of the coin is:
(a) 2.07 mm
(b) 2.70 mm
(c) 2.35 mm
(d) 2.14 mm
- Q20.** The radius of a uniform wire is measured as $r = 0.024 \text{ cm}$. The value of π taken in calculation is 3.1416. The cross-sectional area $A = \pi r^2$ of the wire to correct significant figures is:
(a) $1.8 \times 10^{-3} \text{ cm}^2$
(b) $1.81 \times 10^{-3} \text{ cm}^2$
(c) $1.809 \times 10^{-3} \text{ cm}^2$
(d) $1.8096 \times 10^{-3} \text{ cm}^2$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(c)	6	(c)	11	(c)	16	(b)
2	(b)	7	(d)	12	(a)	17	(a)
3	(a)	8	(a)	13	(b)	18	(a)
4	(a)	9	(a)	14	(c)	19	(b)
5	(c)	10	(a)	15	(c)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (c)

Given: 1 MSD = 1 mm.

$$20 \text{ VSD} = 16 \text{ MSD} \Rightarrow 1 \text{ VSD} = \frac{16}{20} \text{ MSD} = 0.8 \text{ mm.}$$

Least Count:

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = 1 \text{ mm} - 0.8 \text{ mm} = 0.2 \text{ mm}$$

Sol 2. Correct Option: (b)

$$\text{Least count LC} = \frac{\text{Pitch}}{\text{Number of divisions}} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm.}$$

Since the zero of the circular scale lies ahead of the reference line (45th division coinciding with reference mark), it is a negative zero error:

$$\text{Zero Error} = -(50 - 45) \times \text{LC} = -5 \times 0.01 \text{ mm} = -0.05 \text{ mm}$$

Sol 3. Correct Option: (a)

- In 0.003020: Leading zeroes are non-significant; zeroes between non-zero digits and trailing zeroes after decimal are significant $\Rightarrow$ 4 significant figures (3, 0, 2, 0).

- In 4.500×10^3 : The coefficient has trailing zeroes with decimal $\Rightarrow$ 4 significant figures (4, 5, 0, 0).

- In 200.0: Trailing zeroes with explicit decimal point are significant $\Rightarrow$ 4 significant figures.

Sol 4. Correct Option: (a)

$$\text{Given: } N \text{ VSD} = (N - 1) \text{ MSD} \Rightarrow 1 \text{ VSD} = \frac{N - 1}{N} \text{ MSD.}$$

Least count:

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = \left(1 - \frac{N - 1}{N}\right) \text{ MSD} = \frac{1}{N} \text{ MSD}$$

Sol 5. Correct Option: (c)

$$\text{LC} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm.}$$

Observed Reading:

$$\text{MSR} + (\text{CSR} \times \text{LC}) = 2 \text{ mm} + (52 \times 0.01 \text{ mm}) = 2.52 \text{ mm}$$

Corrected Reading:

$$\text{Observed Reading} - (\text{Zero Error}) = 2.52 \text{ mm} - (-0.03 \text{ mm}) = 2.55 \text{ mm}$$

Sol 6. Correct Option: (c)

$$4.28 - 0.125 + 1.2 = 5.355.$$

In addition/subtraction, the final result must retain as many decimal places as the term with the fewest decimal places (1.2 has 1 decimal place).

Rounding 5.355 to 1 decimal place gives 5.4.

Sol 7. Correct Option: (d)

$$\text{Area} = l \times b = 15.42 \text{ cm} \times 4.2 \text{ cm} = 64.764 \text{ cm}^2.$$

In multiplication, the answer should contain significant figures equal to the least precise measurement ($b = 4.2$ has 2 significant figures).

Rounding 64.764 to 2 significant figures yields 65 cm^2 .

Sol 8. Correct Option: (a)

$$\text{Least Count} = \frac{\text{Pitch}}{\text{Number of circular divisions}} = \frac{0.5 \text{ mm}}{100} = 0.005 \text{ mm.}$$

Sol 9. Correct Option: (a)

$$1 \text{ MSD} = 0.5 \text{ mm.}$$

$$50 \text{ VSD} = 49 \text{ MSD} \Rightarrow 1 \text{ VSD} = \frac{49}{50} \text{ MSD.}$$

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = \frac{1}{50} \text{ MSD} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

Sol 10. Correct Option: (a)

$$\text{LC} = \frac{1 \text{ mm}}{100} = 0.01 \text{ mm.}$$

$$\text{Observed Reading} = 3 \text{ mm} + (68 \times 0.01 \text{ mm}) = 3.68 \text{ mm.}$$

$$\text{Correct Reading} = \text{Observed} - \text{Zero Error} = 3.68 \text{ mm} - (+0.04 \text{ mm}) = 3.64 \text{ mm.}$$

Sol 11. Correct Option: (c)

$$V = x^3 \Rightarrow \frac{\Delta V}{V} = 3 \left(\frac{\Delta x}{x} \right).$$

Percentage error:

$$\frac{\Delta V}{V} \times 100 = 3 \left(\frac{0.01}{2.50} \right) \times 100 = 3 \times 0.4\% = 1.2\%$$

Sol 12. Correct Option: (a)

$$1 \text{ MSD} = \frac{1 \text{ cm}}{10} = 1 \text{ mm.}$$

$$25 \text{ VSD} = 11 \text{ MSD} \Rightarrow 1 \text{ VSD} = 1.1 \text{ mm.}$$

For a retrograde vernier scale:

$$LC = |1 \text{ MSD} - 1 \text{ VSD}| = |1 \text{ mm} - 1.1 \text{ mm}| = 0.1 \text{ mm}$$

Sol 13. Correct Option: (b)

Rule for rounding off when preceding digit is followed by 5 (without subsequent non-zero digits):

- For 3.745: Preceding digit 4 is even $\Rightarrow$ unchanged $\Rightarrow$ 3.74.

- For 3.735: Preceding digit 3 is odd $\Rightarrow$ increased by 1 $\Rightarrow$ 3.74.

Sol 14. Correct Option: (c)

$$\rho = \frac{m}{V} = \frac{24.2 \text{ g}}{4.7 \text{ cm}^3} = 5.148936 \dots \text{ g} \cdot \text{cm}^{-3}.$$

m has 3 significant figures and V has 2 significant figures. The final result must have 2 significant figures $\Rightarrow$ $5.1 \text{ g} \cdot \text{cm}^{-3}$.

Sol 15. Correct Option: (c)

MSR = 3.2 cm, VSR = 4, LC = 0.01 cm.

Observed Reading = $3.2 + (4 \times 0.01) = 3.24 \text{ cm}$.

Correct Reading = Observed - Zero Error = $3.24 - (-0.02) = 3.26 \text{ cm}$.

Sol 16. Correct Option: (b)

$$LC = 1 \text{ MSD} - \frac{49}{50} \text{ MSD} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}.$$

$$\text{Number of divisions } n = \frac{\Delta x}{LC} = \frac{0.14 \text{ mm}}{0.01 \text{ mm}} = 14.$$

Sol 17. Correct Option: (a)

$$\text{Pitch} = \frac{\text{Distance moved}}{\text{Number of complete rotations}} = \frac{2.5 \text{ mm}}{5} = 0.5 \text{ mm}.$$

$$\text{Least Count} = \frac{\text{Pitch}}{\text{Circular scale divisions}} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}.$$

Sol 18. Correct Option: (a)

Convert to common exponent:

$$(0.30 \times 10^{-2} \text{ kg}) + (4.50 \times 10^{-2} \text{ kg}) = 4.80 \times 10^{-2} \text{ kg}$$

Both numbers have 2 decimal places when written with exponent 10^{-2} , hence the sum has 2 decimal places $\Rightarrow 4.80 \times 10^{-2} \text{ kg}$.

Sol 19. Correct Option: (b)

1 MSD = 1 mm, 10 VSD = 9 MSD $\Rightarrow$ LC = 0.1 mm.

Thickness = MSR + (VSR $\times$ LC) = $2 \text{ mm} + (7 \times 0.1 \text{ mm}) = 2.70 \text{ mm}$.

Sol 20. Correct Option: (a)

$r = 0.024 \text{ cm}$ has 2 significant figures.

$$A = \pi r^2 = 3.1416 \times (0.024)^2 = 3.1416 \times 0.000576 = 1.80956 \times 10^{-3} \text{ cm}^2.$$

Rounding to 2 significant figures gives $1.8 \times 10^{-3} \text{ cm}^2$.

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